

1 Reducibility

Definition Reduction: transforming one problem into another, such that the solution to the second problem yields the solution to the first one. If, for instance,

```
solve_A(...)
...
...
    solve_B(...)
...
...
return
```

then:

- If B can be solved, then A can be solved.
- Or, negatively, if A is unsolvable, then B is unsolvable.

Notation: $A \leq B$:

- A can be reduced to B .
- A is no harder than B .

Example

```
solve_L_d(...)
...
...
    solve_A_TM(...)
...
...
return
```

So: $L_d \leq A_{TM}$.

1.1 More undecidable problems / languages

- $HALT_{TM} = \{ \langle M, w \rangle \mid M \text{ halts on } w \}$

Theorem 1.1 $HALT_{TM}$ is undecidable.

Proof by reduction of A_{TM}

Assume $HALT_{TM}$ is decidable by some decider R . Then we can construct a decider S for A_{TM} with Algorithm 1.

Algorithm 1 S on $\langle M, w \rangle$

```
Run  $R$  on  $\langle M, w \rangle$ 
if  $R$  rejects (i.e.  $M$  loops on  $w$ ) then
    reject
else
    Run  $M$  on  $\langle w \rangle$ 
    if  $M$  accepts  $w$  then
        accept
    else if  $M$  rejects  $w$  then
        reject
    end if
end if
```

But we know that A_{TM} is undecidable and no such S exists. Then, no such R exists, and $HALT_{TM}$ is undecidable. ■

- $E_{TM} = \{ \langle M \rangle \mid L(M) = \emptyset \}$

Theorem 1.2 E_{TM} is undecidable.

Proof by reduction from A_{TM} .

For a given A_{TM} instance $\langle M, w \rangle$, construct an E_{TM} instance $\langle M_1 \rangle$ such that $L(M_1)$ will be $\emptyset$ or not depending on whether M accepts w or not with Algorithm 2.

Algorithm 2 M_1 on x

```

if  $x \neq w$  then
    reject
else
    Run  $M$  on  $\langle w \rangle$ 
    if  $M$  accepts  $w$  then
        accept
    else if  $M$  rejects  $w$  then
        reject
    end if
end if

```

$$L(M_1) = \begin{cases} \emptyset, & \text{if } w \notin L(M) \\ \{w\}, & \text{if } w \in L(M) \end{cases}$$

Now, assume E_{TM} is decidable by some TM R . We can construct a decider S for A_{TM} using Algorithm 3.

Algorithm 3 S on $\langle M, w \rangle$

```

Construct  $M_1$  from  $M$  and  $w$ 
Run  $R$  on  $\langle M_1 \rangle$ 
if  $R$  accepts then
    reject
else if  $R$  rejects then
    accept
end if

```

We know S does not exist. Therefore E_{TM} is undecidable. ■

- $REGULAR_{TM} = \{ \langle M \rangle \mid L(M) \text{ is a regular language} \}$

Similarly, we will reduce from A_{TM} .

The idea is to construct a new machine M_2 that recognizes a regular language iff M accepts w . modified machine M_2 . Initially, M_2 will recognize the non-regular language $\{0^n 1^n \mid n \geq 0\}$ if M does not accept w , and will recognize the regular language Σ^* if M accepts w .

$M_2 =$ On input x :

1. If x has the form $0^n 1^n$, *accept*.
2. If x does not have this form, run M on input w and accept if M accepts w .

Let R be a decider TM that decides $REGULAR_{TM}$, and let S be a decider for A_{TM} .

Algorithm 4 S on $\langle M, w \rangle$

Construct M_2 from M and w .
 Run R on $\langle M_2 \rangle$
if R accepts **then**
 $accept$
else if R rejects **then**
 $reject$
end if

We know S does not exist. Therefore $REGULAR_{TM}$ is undecidable. ■

- $EQ_{TM} = \{ \langle M_1, M_2 \rangle \mid L(M_1) = L(M_2) \}$

Theorem 1.3 EQ_{TM} is undecidable.

Proof by reduction from E_{TM} .

Assume EQ_{TM} is decidable by R . Then we can have a decider for S as outlined in Algorithm 5.

Algorithm 5 S on $\langle M \rangle$

Construct a TM M_3 that rejects everything. (i.e. $L(M_3) = \emptyset$)
 Run R on $\langle M, M_3 \rangle$
if R accepts **then**
 $accept$
else if R rejects **then**
 $reject$
end if

We know S does not exist. Therefore EQ_{TM} is undecidable. ■

1.2 Rice's Theorem

For any non-trivial property $\mathcal{P}$ of the language of a TM:

$$\mathcal{P}_{TM} = \{ \langle M \rangle \mid L(M) \text{ has property } \mathcal{P} \}$$

is undecidable.

$$\left. \begin{array}{l} L(M) \subseteq \Sigma^* \\ L(M) \supseteq \emptyset \end{array} \right\} \text{ Trivial properties}$$

Example

- $REGULAR_{TM} = \{ \langle M \rangle \mid L(M) \text{ is a regular language} \}$ is undecidable.
- $CF_{TM} = \{ \langle M \rangle \mid L(M) \text{ is a context free language} \}$ is undecidable.